GoodBelly: Using Statistics to Justify the Marketing Expense

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Caroline Dickerson faces a business challenge: GoodBelly's senior management is doubting if the in-store demo program and endcap display competition truly boost sales sufficiently to warrant their expenses. The firm has restricted marketing resources, and if these initiatives fail to show effectiveness, their funding may be reduced. Marty Wellbeing, the marketing manager, instructed Caroline to examine the latest sales and promotion data to: evaluate the effect of the two promotional initiatives on sales, assess whether any increase in sales is statistically meaningful and sustainable, and offer recommendations supported by evidence that can convince doubtful executives, particularly the CFO, to continue funding. To sum up, she needs to justify her market spending to avoid a budget cut.

The sales records from GoodBelly products at 126 Whole Foods stores over ten weeks that are part of our dataset provide a good ground for examining promotional effectiveness, sales drivers. Weekly sales volume is the dependent variable with independent variables consisting of product price, the presence and timing of in-store demos, the number of endcap displays, lagged demo effects, and store location characteristics like size, region, and foot traffic potential. Through data cleaning, I made sure that all stores and time periods were consistent: missing values were filled using prior week trends or were excluded, store IDs and dates were reformatted, and duplicates were removed. Demo variables were changed into dummy variables where "no demo" was the base case, which allowed coefficients to express the incremental effect on sales in relation to weeks without a demo.

Methodology:

Initial Model:

Units Soldi = β0 + β1 · Pricei + β2 · SalesRepi + β3 · Endcapi + β4 · Demoi + β5 · Mid(1-3)i + β6 · Long(4+)i + β7 · SKUsi + β8 · Naturali + β9 · Fitness + ϵi

Explanation:

- β0: Intercept
- B1 to β9: Regression coefficients representing the effect of each variable on weekly sales
- ϵi: Error term

Before fitting the model, I first checked for multicollinearity to see if any variables were highly correlated. All correlations were below 0.7 (Appendix 1), so no variables were removed at this stage.

I then ran the regression analysis, and the F-test was highly significant (Appendix 2), indicating that the model provides a good explanation of weekly sales.

Next, I examined the p-values of each variable. A high p-value indicates that the variable does not have a significant impact on sales. I found that the Natural and Fitness variables had p-values above our significance threshold of 0.05. (Appendix 3) Therefore, I first removed Natural, which had the largest p-value, and reran the regression to check whether the p-values of other variables changed. The results showed that Fitness still exceeded our 0.05 threshold, so it was also removed.

After removing these two variables, all remaining variables had a statistically significant impact on sales. (Appendix 4) I then checked for homoscedasticity, and none of the residual plots showed consistent variance.

Final Model:

Units Soldi = β0 + β1 · Pricei + β2 · SalesRepi + β3 · Endcapi + β4 · Demoi + β5 · Mid(1-3)i + β6 · Long(4+)i + β7 · SKUsi + ϵi

Therefore, I can conclude that price, sales representatives, endcap promotions, in-store demos, and product variety all have a significant impact on GoodBelly's weekly sales.

Our regression model used to produce these suggestions confidently explains 62% of the weekly sales variation, which is a strong fit for retail and CPG marketing analysis. The results show clear, measurable impacts from several key factors. Price sensitivity is substantial, a $1 increase in price results in approximately 24 fewer units sold per week, meaning any price changes should be made cautiously and ideally paired with promotions. Sales representatives provide a consistent benefit, increasing weekly sales by about 80 units, likely due to improved merchandising and stronger store relationships with customers. Expanding SKU variety also yields a modest but steady sales boost.

Among promotional tactics, endcap displays deliver the largest short-term sales lift, driving approximately 305 extra units per week. This suggests that expanding endcap placements could yield a high return on investment. In-store demos also show a significant immediate impact, increasing sales by roughly 120 units in the week they occur. Importantly, the effect of these demos stays, sales remain elevated by about 68 units in the 1-3 weeks following the event and still maintain a 49-unit boost even 4+ weeks later.

These findings counter concerns raised in the initial assumption that demos may only provide a short-lived sales spike. Instead, the data demonstrates that demos not only generate immediate gains but also sustain higher sales over several weeks, providing lasting value. Together, demos and endcaps represent highly effective promotional levers, while sales reps and SKU variety offer consistent support to overall performance. Given these results, maintaining and even expanding these initiatives, particularly endcaps and demos would be a data-backed strategy to maximize sales growth.

I propose the expansion of endcap displays as they provide the best short-term sales lift-about 305 extra units per week- and are the best ROI opportunity. Endcaps, which are placed at both ends of the aisles, have a high shopper visibility feature and thus increase the possibility of impulse buying by customers, especially in busy stores. The cost of renting and stocking this space is relatively low compared to the extra sales that one would get, therefore endcaps are a cost efficient product promotion tool. In order to double the effect, add endcap placements to them in these times of prime sales times, like the holidays, or seasonal events, and possibly the targeted promotions to give a boost to performance even would be placed should be prioritized.

In store demos should also be kept and targeted depending on their immediate effect of increasing sales by 107 units and continuing to gain units by 68 in the 1-3 weeks, 49 units at 4+ weeks countering the initial assumption that their effect is short lived. The majority of these should be markup till to the highest possible in the most visited stores test this assertion. Annual sales representatives at their consistent levels due to their 80 unit contribution, weekly support merchandising quality and promotional execution control. Also, in stores with diverse SKUs and active campaigns, salesquelling should be focused on value-added displays. All things considered, in increasing expenditure to data-back endcaps and demos, ai, retaining sales reps, and sku variety would be the right path to optimizing sales growth.

The analysis being based on a shorter 10 week dataset limits the chance to make more applicable and scalable observations. With this comes the major downside, no long term patterns or seasonal trends can be made note of that impacts the final results. To overcome this, extending the period of observation to multiple seasons or a full year would help to notice these overall impacts. Although the model shows signs of statistical relationships between variables, this however showcases correlations rather than causations. Taking a look at an example, stores with more endcaps may also have higher foot traffic overall which could contribute to higher sales separate to that of promotions. Other important variables like advertising, weather, economic shift, and so on were not noticed when analyzing the current dataset. Including these in future efforts could aid in the model's accuracy and provide a more encompassing understanding of the drivers behind GoodBelly's sales performance.

Our analysis showcases how GoodBelly's continuous investment in their endcaps and in-store demos brings forth noticeable lifts in sales. The model points out about 62% of weekly sales variation where endcaps take part in the largest short term lift of 305 units per week and demos contributing to rapid increases of 120 units. Sales representatives and SKU variety also leaves a positive impact while higher pricing leaves a negative impact on sales. These results highlight the previous challenges about the demo program having a lasting impact past a singular week. Looking ahead, GoodBelly should hone in on maintaining and expanding their high impact programs where they can focus on supportive customer service and encourage promotional strategies. Taking it a step further, analyzing other retail channels over a longer period of time will provide the necessary basis to target and maximize return on their investment.

Appendix

- Appendix 1 - Correlation Matrix

	Units Sold	Price	Sales Rep	Endcap	Demo	Mid(1-3)	Long(4+)	SKUs	Natural	Fitness
Units Sold	1									
Price	-0.018953	1								
Sales Rep	0.449386	0.328792	1							
Endcap	0.593264	-0.067413	0.051886	1						
Demo	0.281685	0.018179	0.109325	0.067265	1					
Mid(1-3)	0.303700	-0.113623	0.175925	0.089343	-0.068013	1				
Long(4+)	0.098632	-0.011364	0.079004	0.022944	-0.041076	-0.070374	1			
SKUs	0.314079	0.371446	0.592594	0.116253	0.057851	0.081550	-0.022578	1		
Natural	0.062687	-0.005356	0.089318	0.081884	0.005863	0.004462	0.020081	-0.028800	1	
Fitness	-0.031728	0.071591	0.057511	-0.059635	0.013149	0.013193	-0.025355	0.085936	-0.085627	1

- Appendix 2 - The ANOVA table

  It shows whether the overall regression model significantly explains sales variation.

ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	9	10646967	1182996.	253.7557	7.34678636572681E-285
Residual	1376	6414842.	4661.949		
Total	1385	17061810			

- Appendix 3 - Regression Output Table (Initial Model)

	Coefficient	*andard Err*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0*	*pper 95.0%*
Intercept	254.1084	19.90939	12.76324	0	215.0524	293.1645	215.0524	293.1645
Price	-23.92572	4.411941	-5.422947	0.000000	-32.58058	-15.27086	-32.58058	-15.27086
Sales Rep	80.99621	4.828882	16.77328	0	71.52344	90.46898	71.52344	90.46898
Endcap	305.1874	9.806310	31.12154	0	285.9505	324.4244	285.9505	324.4244
Demo	120.4670	9.077711	13.27064	0	102.6594	138.2747	102.6594	138.2747
Mid(1-3)	68.02190	5.928447	11.47381	0	56.39213	79.65167	56.39213	79.65167
Long(4+)	48.48045	10.07380	4.812527	0.000001	28.71877	68.24212	28.71877	68.24212
SKUs	3.664034	1.611883	2.273137	0.023171	0.502018	6.826049	0.502018	6.826049
Natural	-2.034806	1.913279	-1.063517	0.287733	-5.788068	1.718454	-5.788068	1.718454
Fitness	-1.619108	1.163867	-1.391144	0.164406	-3.902255	0.664039	-3.902255	0.664039

- Appendix 4 - Regression Output Table (Model after Removing Fitness & Natural Variables)

	Coefficient	andard Err	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0	pper 95.0%
Intercept	247.8350	19.41394	12.76582	0	209.7509	285.9191	209.7509	285.9191
Price	-24.11378	4.410340	-5.467555	0.000000	-32.76548	-15.46207	-32.76548	-15.46207
Sales Rep	80.30308	4.785924	16.77901	0	70.91459	89.69156	70.91459	89.69156
Endcap	305.1534	9.751753	31.29216	0	286.0235	324.2833	286.0235	324.2833
Demo	120.4304	9.079105	13.26457	0	102.6201	138.2408	102.6201	138.2408
Mid(1-3)	68.03948	5.927886	11.47786	0	56.41083	79.66814	56.41083	79.66814
Long(4+)	48.77128	10.07491	4.840864	0.000001	29.00746	68.53510	29.00746	68.53510
SKUs	3.714193	1.601077	2.319808	0.020496	0.573380	6.855006	0.573380	6.855006